Hello Crystal

Heuristic Learning over CrystalRL

a transparent agent that evolves itself — and adapts to every investor

Ivan Pavliuk

July 2026 · a pitch for Joseph Lynch

The dream — and the two walls it always hits

An investment agent that **reads its own diagnostics**, **rewrites its own machinery**, and **adapts itself to each person** — "I can lose 10%, I have 5 years, I need a third more" — without a human re-engineering it for every user.

Why self-evolving agents usually cannot be trusted:

- **Wall 1 — you cannot read the agent.** A black box that changes itself is evolution without eyes: you learn what it became only after it costs money.
- **Wall 2 — you cannot trust the judge.** In markets the evaluator is a backtest: noisy, leaky, and gameable by the very optimization it judges.

Analogy: studying a brain you can neither image nor stimulate, with behavioral scores a subject can fake. Neuroscience solved this with imaging, interventions, and controls — so do we.

Our thesis, in one sentence

An agent may evolve itself **exactly as far as it can be read**.

- **CrystalRL** — the transparent agent: its policy is a **table**, its memory is **named probabilities you can write to**, its objective is the **user's own contract**. Interpretability is not documentation here — it is the load-bearing wall.
- **Heuristic Learning (HL)** — the LLM coding agent on top: it proposes changes to the machinery and **tailors the system to each investor**; experimental controls decide what survives.
- The honesty machinery (calibration gates, refusal, pre-registration) is the **safety rail** that lets the first two run at full speed.

Transparency is what makes self-evolution safe; the coding agent is what makes it cheap.

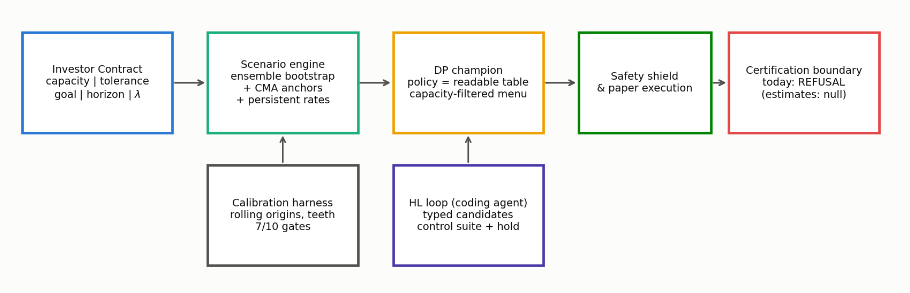
What are you getting from that?

Tell it back in your own words — the fastest way to sync two brains.

the check-understanding habit, borrowed from J. Eisner, *How to work with a professor*

`https://www.cs.jhu.edu/~jason/advice/how-to-work-with-a-professor.html`

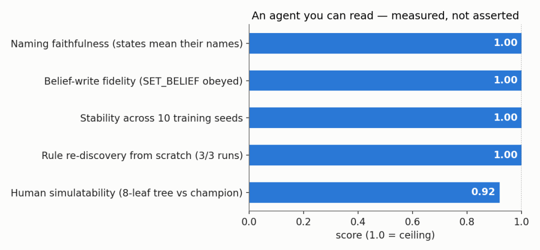
The system in one picture



- **CrystalRL core**: contract → readable policy → execution *a subject you can image AND stimulate*
- **HL loop**: the coding agent proposes; controls decide *sham / dose / replication*
- **Safety rail**: self-calibration + a certification boundary that refuses until evidence matures *publishing the nulls*

CrystalRL — an agent you can read *and* write

- The champion policy is a **table**: *"behind the goal → full allowed equity; at the goal → lock it in"* — an 8-leaf tree reproduces it at **0.92**. *a receptive-field atlas, not electrode guesswork*
- Its memory is **named**: `SET_BELIEF(bear)=0.8` is a **command the policy must obey** — write fidelity **100%**; a ±2-nat nudge slides behavior between named codes. *optogenetics for a policy*
- Transparency is **measured**: the audit puts today's substrate **at ceiling** (right).



Forcing a short story loses **0.0** accuracy (MDL deficit).

The honest frontier: on harder substrates legibility is *not* free — a **certified** return↔legibility tension. Exactly the open research seat.

Heuristic Learning — the coding agent on top

The LLM agent reads the logbook and diagnostics, then proposes **typed changes to the machinery**: new action menus, new scenario members, new solver knobs — **unlimited imagination, zero authority**.

- **matched-random** — *sham stimulation*
- **wrong-direction** — *inverted stimulus*
- **half-dose** — *dose–response*
- **out-of-time** — *another lab*
- **one hold-out read** — *read once*

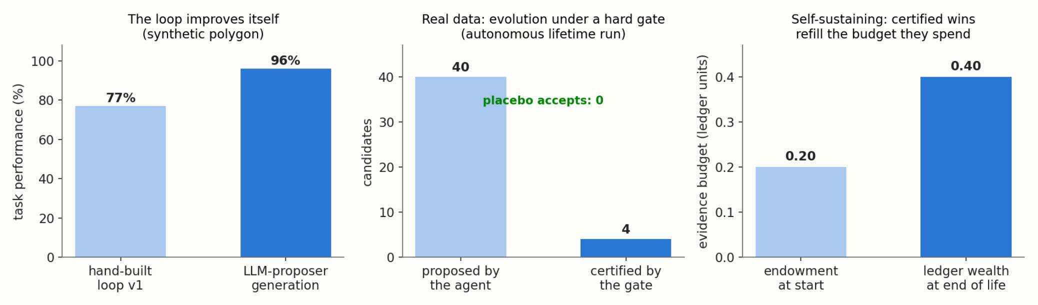
Why the paranoia? FunSearch-style loops trust an *exact* evaluator. Markets are the opposite regime: the judge is a backtest — noisy, leaky, gameable.

So the rule is

When the judge cannot be trusted, **controls decide, not scores**.

And because the *agent* (CrystalRL) is readable, every accepted change can be inspected as a diff on a table or a named belief — evolution with eyes.

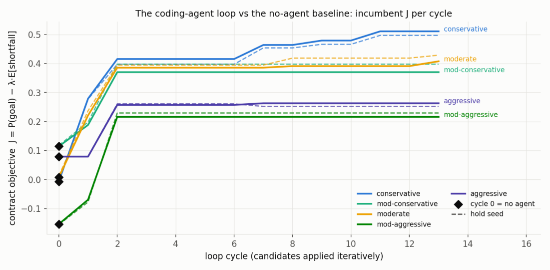
The loop's track record: it already evolves itself



- Synthetic polygon: the LLM-proposer generation lifted the loop's own skill **77% → 96%**.
- First **certified real-data policy**: 7 accepts, **0** placebo accepts — and the result is a *readable sentence*: "$P(\text{bear}) > 0.66 \rightarrow$ exposure 0.74."
- Fully **autonomous lifetime run**: 40 proposals → 4 certified refinements; the evidence budget **grew 0.20 → 0.40** — certified wins refill what experiments spend.
- Red-teamed and broken twice → fixed; the fix verified by the *attacker's own script*.

Adaptation: the loop tailors the system to each investor

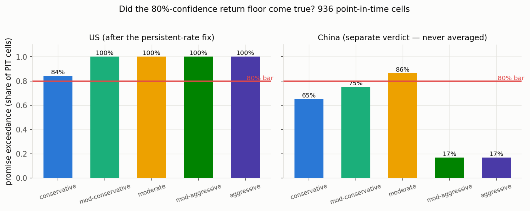
- Every user is a **contract**: capacity $\neq$ tolerance $\neq$ goal *(pain threshold vs pain tolerance)* — and capacity is a hard filter the evolution itself may never relax.
- Infeasible goal? The system **refuses** and offers honest levers: time, contributions, a smaller goal — never more risk.
- Then the loop runs **per profile**: five investors, one machinery, **five differently-evolved menus**.



Per-profile learning curves vs. the no-agent cycle-0 baseline; solid = development, dashed = held-out.

65 cycles, 15 accepted, all 15 **hold-confirmed** — conservative profile: goal probability **1%** $\rightarrow$ **52%** at an unchanged risk budget.

The safety rail holds (so evolution can run)



The promise backtest: quote at every past date, replay the same policy on what actually happened.

- **936 point-in-time cells, two markets**: US **5/5 profiles ≥80%** exceedance; risk (drawdown) promises **100% of all cells, both markets**.
- China: 1/5 — separate verdict, never averaged; all three failures diagnosed, fixes queued.
- Engine self-calibration: 7/10 gates green → every number still wears the UNCALIBRATED label; certification **refuses by design**, with dated unlocks (2027 / 2029).

The replication-crisis lessons — pre-registration, publishing nulls, no averaging across species — implemented as code.

What did not work — the graveyard that makes survivors believable

- A famous valuation indicator (**CAPE**): made forecasts worse → falsified, reverted.
- The **PPO challenger**: loses to the readable table → reported, not hidden.
- A **Nasdaq-tracking sleeve** dominated every Chinese profile → red-flagged *by the agent itself*: its edge is one untestable sample era.
- The **conservative promise failed its first backtest** (48.5% vs 80%) → the diagnosis became a structural fix: 48.5% → 84.2%, repairing *exactly* the diagnosed cells.
- On hard substrates, **legibility is not free**: a certified return↔legibility tension — the open frontier of the interpretability track.

A system that demonstrably **kills its own ideas** is the one whose surviving claims — and whose *self-modifications* — you can trust.

Why this becomes a product (the investor view)

- **Personalization at software cost**: today a tailored strategy needs a human quant per client segment. Here **the coding agent does the tailoring** — five profiles already served by one machinery, menus evolved per user, every change certified.
- **A transparent agent is an auditable agent**: policies are tables, beliefs are named and writable — **an audit standard competitors do not have**, in a market where regulators demand suitability and explainability.
- **Self-evolution compounds**: the autonomous run already *funds its own experiments* (evidence budget $0.20 \to 0.40$) — the machine gets better while we sleep, under gates.
- **Already live**: one certified defensive rule on a paper track — on 629 untouched days: **Sharpe 1.46 vs 1.39**, **max drawdown −12.9% vs −15.5%**.
- **Credibility you can date**: 80 forecasts locked before realization; scoring reads **2027 / 2029**. Auditable, not marketing.

Your part: a six-month research program, not a task list

The frame

You are the **neuroscientist of an artificial subject**: read–write access to every "synapse," behavior reproducible by seed, interventions free and ethical — and the subject *changes itself*, so interpretability must **keep up with evolution**.

1. **Month 1 — the measuring instruments**: simulatability psychophysics (can a human predict the policy in unseen cells?); harden the ceiling metrics — do they survive new seeds, substrates, profiles?
2. **Months 2–3 — the atlas**: an *ethogram* of behavioral motifs across profiles; belief faithfulness (does $P(\mathrm{bear})$ encode what its name claims?); the write-protocol (`SET_BELIEF` as optogenetics, mapped).
3. **Months 3–5 — your own loops**: run HL loops with **your own coding agent** to push the certified return↔legibility frontier — where legibility is *not* free.
4. **Months 5–6+ — the explanation gate**: every "why" the product tells a user passes faithfulness tests *you* own; standing red-team seat *(dead salmon)*; co-author *Hello Crystal* + lead the methods paper (*"an ethogram of an artificial investor"*).

How we will work together

Weekly sync (Sunday): online meeting *or* a recorded video.

Agenda sent 1–2 days in advance, four points:

1. what you have worked on since we last met,
2. what you are struggling with,
3. what you need help with,
4. what your goals are for next time.

Working principles:

- **Write the paper first** — the draft exists; `\joseph{}` todos are waiting for you.
- **Shared repo + honest logbook** — every run logged, nulls first; be ready to hand off at any time.
- **Lead with the null** — a negative result is a result.
- **Ask early** — asking beats a week of guessing.

format per J. Eisner, *How to work with a professor* (“working with others”) + M. Dredze, *How to be a successful PhD student*

`https://www.cs.jhu.edu/~jason/advice/how-to-work-with-a-professor.html`

The ask

Own the **interpretability track** — the load-bearing wall of a self-evolving agent — as **co-author** of *Hello Crystal*.

By next Sunday:

- run the sandbox and read the top-5 logbook entries (an afternoon);
- tell me which **phase of the six-month program** you would reshape — it is a proposal, not a syllabus;
- bring **three objections** — breaking our explanations is the job.

Hello, Crystal.

The subject is waiting to be studied — and it will not sit still.